

ZHANGIZTOBE, KazakhstanWMO: **363970**Lat: **49.217N**Lon: **81.217E**Elev: **1493**StdP: **13.92**Time Zone: **6.00 (E06)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-25.4	-37.1	0.7	-31.8	-30.6	1.0	-25.3	54.5	18.2	53.2	17.0	1.9	190	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	91.8	63.6	88.1	62.9	85.1	61.9	66.7	83.2	65.4	81.5	64.2	79.4	7.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	87.8	70.9	60.2	82.2	70.4	58.5	77.5	69.7	32.3	83.2	31.1	81.9	30.2	79.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	44.3	38.8	35.5	DB	-32.0	96.6	8.4	3.1	-38.1	98.8	-43.0	100.6	-47.7	102.4	-53.8	104.6
(5)				WB	-31.4	70.3	8.3	2.2	-37.4	71.9	-42.3	73.2	-46.9	74.5	-53.0	76.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	3.6	7.5	22.7	45.5	57.1	68.3	70.5	67.5	56.0	42.2	24.5	10.3
	DBStd	26.30	14.26	13.79	13.07	10.30	8.56	6.93	6.05	6.83	9.64	9.71	14.16	13.54
	HDD50	6028	1440	1190	847	197	33	0	0	0	48	278	765	1230
	HDD65	9714	1905	1610	1312	585	267	48	20	52	298	707	1215	1695
	CDD50	2306	0	0	0	62	253	549	637	542	227	36	0	0
	CDD65	518	0	0	0	0	22	147	192	129	28	0	0	0
	CDH74	5967	0	0	0	22	377	1560	1973	1587	443	5	0	0
	CDH80	2166	0	0	0	1	101	547	756	617	144	0	0	0
Wind	WSAvg	9.8	15.8	14.8	11.4	7.5	7.6	6.4	6.2	5.8	6.7	8.5	12.0	15.6
Precipitation	PrecAvg	12.4	0.6	0.6	0.7	0.9	1.1	1.6	2.0	0.9	0.7	1.0	1.2	0.9
	PrecMax	17.4	1.1	1.2	1.8	3.0	2.7	4.2	5.9	3.4	1.8	2.5	2.0	1.9
	PrecMin	7.6	0.1	0.2	0.0	0.2	0.2	0.5	0.1	0.1	0.0	0.2	0.1	0.2
	PrecStd	2.2	0.3	0.3	0.5	0.6	0.6	0.9	1.2	0.7	0.5	0.6	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	35.2	54.8	77.4	88.8	93.4	96.6	96.3	89.7	73.7	55.9	35.8
		MCWB	28.0	32.0	44.1	57.1	62.5	62.8	66.9	63.8	61.6	52.4	44.1	31.4
	2%	DB	26.4	31.3	48.6	71.7	82.2	89.1	91.2	91.0	84.4	67.4	49.5	32.9
		MCWB	23.8	28.4	41.1	54.0	60.6	63.4	64.5	63.6	58.4	49.7	41.0	30.0
	5%	DB	23.7	28.0	43.4	67.3	77.9	85.9	87.4	86.6	79.0	62.0	44.8	29.2
		MCWB	21.2	25.3	37.9	51.9	57.9	62.6	63.6	61.7	56.4	47.4	38.6	26.7
	10%	DB	20.9	24.8	39.0	62.8	74.0	82.8	84.2	82.7	73.9	57.0	40.3	26.1
		MCWB	18.6	22.3	34.9	49.2	56.1	61.8	62.6	60.5	54.2	45.4	35.5	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.5	32.7	45.8	59.2	65.8	67.8	69.3	67.7	62.5	54.1	46.2	32.6
		MCDB	31.0	34.6	53.3	75.3	83.3	83.2	84.2	86.5	87.0	67.8	53.2	35.2
	2%	WB	24.1	28.8	41.4	55.5	62.1	66.3	67.1	65.2	59.7	51.0	42.1	30.1
		MCDB	26.1	31.0	47.6	69.1	77.8	82.3	82.8	84.1	80.6	64.1	47.9	32.6
	5%	WB	21.4	25.4	37.9	53.0	59.6	64.9	65.7	63.7	57.5	48.6	38.6	26.8
		MCDB	23.6	27.7	43.1	65.1	74.0	80.2	81.1	80.9	75.7	60.2	44.6	29.2
	10%	WB	18.6	22.3	34.8	50.5	57.5	63.4	64.5	62.2	55.4	46.3	35.6	23.9
		MCDB	20.8	24.7	38.7	60.9	71.0	78.2	79.0	78.1	71.6	56.3	40.2	25.9
Mean Daily Temperature Range	5% DB	MDBR	12.7	14.5	14.9	21.0	23.5	21.7	20.9	23.7	24.1	18.5	12.9	11.9
		MCDBR	12.2	13.7	18.9	28.1	29.2	26.4	26.6	29.4	31.1	26.5	17.2	12.5
		MCWBR	11.1	12.2	13.0	14.8	13.8	9.1	8.6	10.3	13.7	13.7	10.9	11.2
	5% WB	MCDBR	12.9	13.6	18.1	24.5	25.5	22.2	21.7	24.8	27.9	23.6	16.6	12.4
		MCWBR	11.9	12.5	12.9	14.1	13.2	8.9	8.6	10.1	12.9	13.2	11.5	11.5
Clear-Sky Solar Irradiance	taub	0.291	0.326	0.380	0.415	0.407	0.408	0.426	0.392	0.350	0.351	0.331	0.298	
	taud	2.075	2.008	2.015	2.104	2.188	2.261	2.223	2.294	2.357	2.253	2.142	2.049	
	Ebn at Noon	231	250	257	262	271	271	263	267	268	242	213	206	
	Edh at Noon	28	38	45	46	44	41	43	38	32	30	26	25	
All-Sky Solar Radiation	RadAvg	529	886	1297	1573	2023	2128	1966	1821	1370	790	442	385	
	RadStd	26	43	101	122	120	131	130	95	110	74	38	34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days		
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